

**Test Report No. F690501/LF-CTSGP05-1125**

Date: October 24, 2005

Page 1 of 2

**To: SAMSUNG ELEC CO., LTD.(ONYANG)**San#74, Buksu-ri  
Baeban-myeon  
Asan-city  
Chungnam 336-851  
korea

The following merchandise was submitted and identified by the client as :

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|                             |                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------|
| <b>Commodity</b>            | : 400FBGA                                                                             |
| <b>SGS File No.</b>         | : GP05-1125                                                                           |
| <b>Received Date</b>        | : October 17, 2005                                                                    |
| <b>Test Performing Date</b> | : October 18, 2005                                                                    |
| <b>Test Performed</b>       | : SGS Testing Korea tested the sample(s) selected by applicant with following results |
| <b>Test Results</b>         | : For further details, please refer to following page(s)                              |

**SGS Testing Korea Co. Ltd.**

Jeff Jang / Technical Mgr

Jason Han / Lab Director

**Test Report No. F690501/LF-CTSGP05-1125**

Date: October 24, 2005

Page 2 of 2

Sample No. : GP05-1125.001

Sample Description : 400FBGA

Style/Item No. : MMSP-2

**Heavy Metals**

| Test Items                  | Unit  | Test Method          | MDL | Results |
|-----------------------------|-------|----------------------|-----|---------|
| Cadmium(Cd)                 | mg/kg | USEPA 3050B, ICP-AES | 0.5 | N.D.    |
| Lead (Pb)                   | mg/kg | USEPA 3050B, ICP-AES | 5   | 21.6    |
| Mercury (Hg)                | mg/kg | USEPA 3052, ICP-AES  | 2   | N.D.    |
| Hexavalent Chromium (Cr VI) | mg/kg | USEPA 3060A, UV-vis  | 1   | N.D.    |

**Flame Retardants-PBBs/PBDEs**

| Test Items               | Unit  | Test Method        | MDL | Results |
|--------------------------|-------|--------------------|-----|---------|
| Monobromobiphenyl        | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Dibromobiphenyl          | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Tribromobiphenyl         | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Tetrabromobiphenyl       | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Heptabromobiphenyl       | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Hexabromobiphenyl        | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Pentabromobiphenyl       | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Octabromobiphenyl        | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Nonabromobiphenyl        | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Decabromobiphenyl        | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Monobromobiphenyl ether  | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Dibromobiphenyl ether    | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Tribromobiphenyl ether   | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Tetrabromobiphenyl ether | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Heptabromobiphenyl ether | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Hexabromobiphenyl ether  | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Pentabromobiphenyl ether | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Octabromobiphenyl ether  | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Nonabromobiphenyl ether  | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |
| Decabromobiphenyl ether  | mg/kg | USEPA 3540C, GC/MS | 5   | N.D.    |

\*\*\* End \*\*\*

NOTE: N.D. = Not detected.(<MDL)  
ppm = mg/kg  
MDL = Method Detection Limit  
"- " = No Regulation  
\*\* = Qualitative analysis (No Unit)  
Negative = Undetectable / Positive = Detectable